

2025 ReviewStream

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All right, it is December 18th, 2025. This is going to be a brief 2025 review stream going over the 72 videos of 2025. And just to review, recap, make a kind of concentrated, distilled final end product, taking a quick look back, seeing if any AI perplexity tools can help us get another look at the videos, looking at anyone's comments for who's watching live, seeing if we have any memories, any reflections, any trends that were arising during the year or in relationship with previous years and looking towards what could be in 2026. So let's collapse all these and we'll go from the top. Okay, we'll just reorder things as we go. There was uh well first just at the level of the series and the KOD page that I'm at is in the video description if you look um our videos our streams are sorted into different series and they are reflected upon here. So, one math art stream, 35 guest streams, that's the most general kind of oneoff. Four quarterly roundtables, the last one coming tomorrow. One or stream, two model streams, three math streams, this review stream, 10 textbook groups, [snorts] three very long videos for the applied active inference symposium, and 12 active inference streams with some sort of coding and various works. So let's just jump in. In the first we have feedback welcome with Andrew Penland. And let's just see if Perplexity can summarize this video. This was a fun video because it involved in contrast with pretty much any other video a a very specifically visual component using OBS for feedback kind of like that. So that's fun to uh play play with the with the format and talk about feedback, talk about strange loops, all these different kinds of things that come up in the course of math but also get reflected on in the art. Uh we had three quarterly roundts so far and the fourth one will be happening tomorrow. These are organizational update videos moving from the the practice of organizations into the theory of organization. or stream 9 with Manuel Hoffman was on the value of open source software and here is what comes from that perplexity on feedback welcome. Interesting. This was just a generic, but in other cases, Perplexity is able to have some access to live streams, but appears to have some issues. But on this one, it came out just fine. It's all good. We won't really work with the perplexity. We'll just see what reflections people manually have. Um, this org stream was awesome. I think it's very important to explore this question of of the the value and the

support of open-source software. And I think it provides a great starting point for talking about how do we evaluate the value of what is made openly. And you can see their kind of evaluation uh strategies which are the norms in economics and uh talk about other think about other ways to value software. So very cool to see this kind of pragmatism brought and to bring value to what is often easy to dismiss as simply free. In this year there were two model streams earlier in the year in March and April. Model streams are intended for kind of technical walkthroughs on generative models. So we had model stream 17 with the RX infer team and with Frasier and this was a provocative presentation about what is wrong with active inference of course a bit of a Hegelian approach here to then propose their own work but it was a great follow-on discussion and led nicely into a lot of the ongoing work that's happening with RX X and Fur in model stream 18. This was a presentation on distributively robust pre-energy principle for decision-m. Okay, I'm going to update the title in the production document. Just for a context, this this KOD page that I'm referencing is where all of the live stream information is coordinated. The video is open to everybody with the link. You won't have edit access, but that's where the protocol, the procedure for how we schedule and do live streams and who people have brought up to invite, all these kinds of things. And in this video, they presented on their robotics research and about how to deal with robustness of decision-m when there's uncertainty about the robots world. there's of course this review stream, strange loopy as it may be. There were three math streams, 12, 13, and 14. These were some of the most clear memories for me. I I always enjoyed these mouse streams. Uh in 12.1 Ola running uh presented on the the elbowing stein. So evidence lower bound is equivalent essentially to variational free energy. So just uh sorry about that. We should be back. Um, very nice statistical methods covering some of these distributional update rules and and ways to uh go back and forth and to use these particles, inject these particles in as tracers on a vector field. Mathstream 13 disco pi monoidal categories for active inference of interest for those who are exploring the category theory active inference interface and disco pi is a programming package that helps represent uh string diagrams. So here's a string diagram for cooking and Alexi goes into string diagrams for for active inference and uh gives some background on category theory. Then in math stream 14, Jesse von presented on his review paper a concise mathematical description of active inference in discrete time. So [snorts] great concise pedagogical materials. There were more than 10 but I guess only 10 represented here uh textbook groups. This represented some of the closing times probably for this cycle at least that we are working through the 2022 active inference free energy principle in mind, brain and

behavior textbook as we are eagerly looking towards the fundamentals of active inference principles, algorithms and applications. of the free energy principle for engineers by Sanjiv Nam Jooshi coming out in March of next year. So we're getting excited for that textbook group and looking back on these textbook groups. These are always fun there. There's there's so much to this material and to appreciate Andrew Basha for facilitating a bunch of the textbook group meetings and to all the participants who show up. Uh active inference streams. These were streams that that I expose ways to show what I'm working on often in the active inference or in the meta research or in the tool development space. And the year began with some plain text exploration. April agents bring May simulations that was exploring some different agentic systems. the more LLM coordinationbased open manis as well as the probabilistic generative models with RX infer cerebrum began a thread related to linguistics and basian approaches to language noorg as a organizational agent coordination framework generalized notation notation the following three and 14 and this was moving through this domain specific language for generative models and going from plain text to triple play. So from the configured file into the code that's executable the mathematical representation diagrammatic representation natural language there was the golden spike moment for multi-agent trajectory planning where I reverse engineered the example of multi-agent trajectory into this configurable format and then forward engineered the configuration and we met in the middle had a golden spike moment and that is hopefully of interest to people who who want to see that kind of industrial grade scientific integrity for generative models and the sound of uncertainty which was adding some features into the GNN package like audio format rendering directly of generative models not just their outputs like listening to the configuration start here operational metascientific arts and crafts and generative education yesterday capped off a sort of content development, educational and research tool development arc. Uh, and each of these videos have their associated code and so all open source. Now we get to the most variable which will be the symposium and the guest stream. So the applied active inference symposium, the fifth one was in November 2025. [snorts] And here was the public program just for reference. We had three three days live streams, keynote by Carl Fristen, bunch of live streams, some more workshop sections, and a bunch of pre-recorded videos. And we have the timestamp information available for all of them. Those have been added to the videos. So about 40 or 50 presenters really awesome to see and looking forward to the next year's symposium. And then just last section here for this video is looking at the guest streams and let's see here what trends might emerge. We had rhythmic prochy self uh active imprints for

self-organizing multi-modal systems a Bayesian thermodynamic approach to adaptation. This was very on trend I think for 2025 looking for similarities, differences, convergences, synergies with active inference, probabilistic modeling, cognitive modeling and LLM which was uh one of the themes of the year. In January, Ron Edelman presented on cognitive experience design, designing AI systems for trust. Also complimenting I think another part of that trend which is not just at the technical backend side but also in terms of the human computer interface, user interface side. How how do we design how do we think about these concepts like trust? Matt Sims presented on slime mold and philosophy and Matt had presented here's the full video table. Matt had joined us in October 2021 to discuss his paper how to count biological minds, symbiosis and the free energy principle. So classic Dave Douglas, Steven Clet joining as well. Uh and joined us this year to discuss his book on slime mold which was a a book I got the physical version of and read. Nice, quick, concise read. In February, we had the group of Marie Smith presenting on their work on human computer interaction. So here and with Arun who who coordinated this presentation. There's so so many uh so many interesting analyses. I think when we look across all the videos, which which visual representations of active inference of of the agent environments do people use? What's the rhetoric? What's the technical justification for the different kinds of works? What what do people bring up as their uh their justification? What do they highlight as their their challenges and their successes? This group in particular put in a lot of work to making the different stages of their generative models operation very clear. So great presentation guest stream 90.2 with Benjamin Nelson a intern. This was uh I was in a coffee shop and there were some disruptions which I guess was also sort of the theme of the year. Um, and it was very interesting although all over the place and I hope was a useful milestone and marker on Benjamin's journey going into May. AL Gwen and Carloo presented on from contextuality to social constraints and active inference ontology for participatory realism and social change. Those familiar with Advil's work will will know he always brings a very very rich very holistic view to to the technical elements of the work and and extends and connects it into the social sciences and some may remember a different version of this image in the 2023 course on the social sciences gueststream 100.1 palendrome with you Kanazawa on the free energy principle and implications for language learning and education. This was a very memorable stream for me. Uh partially I think it was nighttime could have been could have been morning in Japan. Use style is extremely energetic and dense visual information that really brought together a great number of points and connected it to language learning and education which is something we've all done and/or are

doing. So this was a very fun and relevant presentation into April. Little bit of a busier April. Gueststream 101 with Mosen Jafari and Yulan Lee was on navigating autonomous vehicles on unmarked road with the fusionbased motion prediction and active inference. So this is uh getting into the applications in robotics and autonomous vehicles. and and smart homes. And so uh and also to the more technical theme of similarities and differences with generative AI and active inference. Here we saw the diffusion modeling and the way that the forward and reverse diffusion modeling could be used which came up in some other streams. 4 9 25 2 3 squared 5 squ with Majid Benny who we will look at his some of his previous uh times joining here was a great stream with Dean and Majid always has a lot of uh very interesting philosophical points. He joined. Interesting. Majid was the first guest stream in February 2021. A critical analysis of Marovian monism. Very first guest stream. Oh, total throwbacks. Total throwbacks. And so while what's new and recent is great, it's also really nice to look look back at some of these older presentations, see where things were the paper and then in 2022 did Majid. Yes, Majid worked on Tale of Two Architectures and then in the of course epic Dean infused live stream 47 series on the double play with inactive social cognition and abduction. The abduction thread continued to be developed by Mjid and colleagues. Still in April this year, we had 82.5 with Robert Warden, a institute research fellow on computers meaning and consciousness. And this was part of a series by Warden. He has now presented on seven guest that took us from the requirements equation, threedimensional spatial cognition, projective wave theory of consciousness and assessment of the brain wave hypothesis. That was last year. And then this year in April and October was computers meaning consciousness, unified theory of language and language and human nature. This was a a a rockness roundt discussion and uh had a bunch of guests in April. Tobias Plowman presented on the mind in motion a systems model of real-time cognitive transitions. So this was a cognitive modeling approach that Tobias had developed. April Tanahhiro Tanaguchi joined for collective predictive coding and active inference a new perspective on emergent communication. And let's look at [snorts] He also joined later in guest stream 118 for co-creative learning via Metropolis Hastings interactions and these were both excellent presentations also nighttime for me also fantastic visually comprehensive technically distilled Very nice presentations bringing together the fundamental, the biological, the generative, active inference. And in the second of Tatahhiro's presentations on the channel, he talked more about extending from the collective predictive coding into the Metropolis Hastings naming game and about this sort of third option that is neither supervised nor unsupervised but co-creative. And so that was very very

thoughtprovoking and is a great example of where something can have both a narrow technical contribution to make as well as a broader philosophical point. and and Todd Herro really spanned that gamut. into May. Paulo Deio joined to discuss systematic deviation presented on a a variety of of topics from from the chemical to the societal. Gueststream 106 was with Bangloo and collaborators and they had written this very large review and synthesis paper on foundation agents and presented on each of the authors presented on on different sections giving a a really nice almost textbook like presentation summarizing the the state of the field and uh great place to to learn about agentic and multi-agentic systems. Gestream 107, David Beno, who's one of the IY organizers, presented on computational psychiatry and affective states, some of the modeling work that their group are involved in. Guest stream 108 with Christo K. Thomas next generation artificial intelligence for emergent semantic communication. This though of course I don't play favorites was one of the most memorable presentations uh for me also with with excellent dense visual elements. But getting into this question of applied semantic transfer and semantic information contents which had been a key theme in our learning going all the way back to 2021 with Chris Fields at all and the information flow in a context dependent hierarchical basian inference and Christo presented on this in the mobile networks similar Tata Herro's work in a way regarding multi- aent semantic transfer and just very fascinating discussion about how as our communication networks move from syntax aware bandwidth like how many bits of information can be transferred how much how many hard drives will it fill up into the semantics of information transfer we get into this real cognitive modeling setting where we're talking about belief updating, message passing and you could have uh one gigabyte file that in practical pragmatic terms doesn't provide any information and you could have one bit of Shannon information that provides uh important reduction of uncertainty for a given agent in a moment in a context. So how do we really work with that? And that's where active inference came into play. And so it was a great discussion. Alexander Heming in 109 presented on free energy, attentional narrow narrowing and cognitive rigidity. This was I think a a sort of cousin approach to that of Tobias in that this was a um cognitive modeling, mathematical and computational model that was connected to a lot of day-to-day uh aspects, cognitive systems and uh is just cool to see all these snapshots of work that's ongoing. In June, Duya is Samura joined for triple equivalence for the emergence of biological intelligence and this was an excellent presentation. Takuya had joined previously in the discussions of live stream 51 on canonical canonical neural networks perform active inference. So here was the double play in 2022 was neural network loss

function and the morphism with a variational base. And then in 2025 he took that two to a three went into the triple play and in the triple play brought in not just the neural network and the variational base but the touring machine formulation. So that uh went another exciting technically sophisticated work bringing together some of the major continents really in in the intelligence landscape today the the neural network the basian and the touring. So great and charismatic fellow guest stream 111.1 with Gorgi Gorgif was on modeling and predicting self-organization in dynamic systems out of thermodynamic equilibrium. This was a nice long presentation from professor Gyorg uh involved a variety of theoretical and empirical features of far from equilibrium systems. So we had a great broad scoping discussion guest stream 112 with Hadiv AI on brain like variational inference kind of a classic postto great um background we see the elbow the evidence lower bound negative free energy coming into play highlighting the unifying nature of variational free energy minimization. coming at this from more of a statistical families of distributions style rather than for example the more um styled unifying where here's more of different kinds of cognitive phenomena, optimal control and model evolution and selection, whereas Hadi made a very walkth through of variational inference and the elbow and presented on on this excellent statistical work with Sergio Yakz on basian mechanics of economic choice computational foundations of economic behavior. This was a great stream. Brought together a lot of the history and classics and fundamentals of micro and macroeconomics and got all the way on through the kinds of modeling that Sergio and colleagues are engaged in right now and I know has developed significantly since this presentation as well. Gueststream 114, Chris Fields, Adam Saffron, Maxwell Ramstead, and Carl Fristen was on the inner screens paper. So the inner screen paper related to nested blankets, to quantum active inference and reference frames, and to awareness and consciousness. So, as one could imagine, it brought out a lot of fun, thought-provoking ideas. I remember this late July, early August period where there was kind of a a turn to energy and Alexi Gladstone presented on energy based transformers are scalable learners and thinkers and this was kind of like bringing together energy based models and transforming formers and was a a very sparse, concise, clear, relevant presentation that is uh definitely important for those who are looking at energy based models and about what where where transformers can can go from here. Then taking a little bit more of a physical material perspective, Max Schaer presented on solving the compute crisis with physics-based AS6. Those are application specific integrated computers. And Max was first author on this review and perspective piece. So basically he took us through the paper that he wrote with colleagues for those interested in

unconventional and conventional computing. It was a great paper and obviously with critical economic and pragmatic relevance going into guest stream 117, Robert Stamps presented on active inference and artificial spin ice control processes and state selection. Here's Vladimir and Robert was presenting on the nanomagnet so material substrate and how predictive coding was being implemented on this material nanomagnetic substrate. 118 was Todd Herro's second talk gueststream 119 with Nick Chader and Robert McKay. This was another economics presentation and this one was a little bit more coming from the macroeconomics side. Still very market oriented and some exciting developments in terms of understanding markets from first principles. Gueststream 120 was with Ifidio Emanuel he presented on several projects related to temporality of cognitive systems and the physics of eye movement. Brought in some consciousness topics and so it was a fun and interesting talk. Getting to the last two here. Guest room 121, Mike Wuest, conscious active inference. This brought together some with a very provocative cover slide, might I add, on quantum active inference, less from the information processing and more from the material substrate of quantum or that type of area connected to the cognitive and also the consciousness aspects. Warden guest room 122 was with Andrea H and she presented on wayfinding radical relation scaling without hierarchies models without the meat great neuroscience informed philosophy and always a good experience to listen and learn from how she's thinking through these things. Warden again on 115 was Marcelo Guzman who presented on learning in physical networks. So this one I would say was a great cousin of Robert Stamps in terms of the nanomagnetics. And here's another example of using physical unconventional computing substrates and big discussion topics around well okay what are the areas that they can hot swap in? What are kinds of cognitive or computational functions that these unconventional substrates can immediately be used for? And how do we program? How do we work with them? How do we model them? Gueststream 124, Ty Roachford, who presented on PCT and FE and discussed some of his work in comparing and holding up active inference alongside the perceptual control theory kinds of models and connecting them with category theory. And even just yesterday uh Martin Na and Osman presenting on from Charles Darwin's root brain to Nicola Tesla's 6G worldbrain and XAI native 6G networks similar to Christo Thomas's presentation in the focus on telecommunication networks with a few different flavors brought in that was very interesting and a great power outage disruption. So if anyone has a comment, write it. Otherwise, that will basically end it. Just wanted to touch upon each of them. Show one last kind of distilled artifact of what happened in this year and look towards

next year. So if you have a idea, get in contact, leave a comment and hope these were epistemic andor pragmatic for you. Thanks for watching, for participating along, and see you next time.